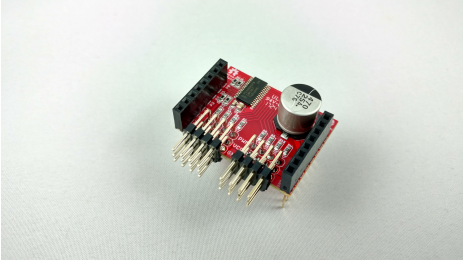
OpenMV® Servo Shield

Datasheet

SKU: OMV-SHIELD-SERVO-V2 / OMV-SHIELD-SERVO-V3

Description

The Servo Shield allows your OpenMV Cam to directly control up to 4 standard hobby servos via P7/P8/P9/P10. Each output is buffered and boosted to 0\u20135 V. Unused outputs can be disconnected via solder-bridge jumpers. The shield features a wide voltage range 6\u201336 V input with reverse voltage and transient surge protection, capable of powering both the servos (up to 5 A) and the OpenMV Cam. Includes a robust 0\u20135 V ADC input on P6 with 4\u201320 mA mode option and a robust 0\u20135 V digital I/O on P10. The V3 updates all I/O pins to bidirectional with 3.3 V to 5 V level shifting, uses lower-profile components, higher-temperature-rated caps, and powers the camera through an ideal diode.



Target Areas

Robotics, pan-tilt systems, gimbal control, actuator control, industrial automation.

Key Features

- **4-channel servo control** — Control up to 4 standard hobby servos via P7/P8/P9/P10. Buffered and boosted to 0–5 V.
- **5 A servo power** — Wide-range regulator provides up to 5 A for servos plus camera power.
- **6–36 V wide-range input** — Reverse voltage and transient surge protection. Outputs 5.6 V on VIN via ideal OR'ing diode.
- **Robust ADC input** — 0–5 V ADC on P6 with overvoltage protection (±36 V tested). Optional 4–20 mA mode via solder bridge.
- **Robust trigger I/O** — 0–5 V digital I/O on P10. Tolerant to reverse voltage, shorts to GND and 12 V.
- **V3 improvements** — Bidirectional I/O with 3.3–5 V level shifting, lower-profile components, ideal diode power.

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1 Features

1.1 Servo Control

Component	Details
Channels	4 servo outputs on P7, P8, P9, P10. Each buffered and level-shifted to 0–5 V.
Servo Power	Wide-range regulator provides up to 5 A for servos.
Disconnectable	Each output can be disconnected via solder-bridge jumper.

1.2 Power Input

Component	Details
Wide-Range Input	6–36 V input with reverse voltage and transient surge protection. Outputs 5.6 V on VIN via ideal OR'ing diode for multi-shield stacking.

1.3 ADC Input

Component	Details
AIN	Robust 0–5 V ADC input, outputs 0–3.3 V on P6. Reverse voltage and overvoltage protection (tested to ±36 V). Optional 4–20 mA mode via 240 Ω solder bridge. Disconnectable.

1.4 Digital I/O

Component	Details
SYN / P10	Robust 0–5 V digital I/O. Tolerant to reverse voltage (-12 V tested), shorts to GND and 12 V. Default: open-drain input on P10. Configurable as 0–5 V digital output. Disconnectable.

1.5 V3 Changes

Component	Details
Bidirectional I/O	All I/O pins (P7/P8/P9/P10) upgraded to bidirectional with 3.3 V to 5 V level shifting (V2 was unidirectional).
Power	Camera powered through ideal diode to reduce heat loss.
Components	Lower-profile components and higher-temperature-rated capacitors.

1.6 Compatibility

Component	Details
Supported Cameras	All OpenMV Cam models.

2 General Info

Dimension	Value
Weight	11 g

3 Company Information

Field	Details
Website	https://openmv.io

4 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/servo-shield
Documentation	https://docs.openmv.io

5 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.